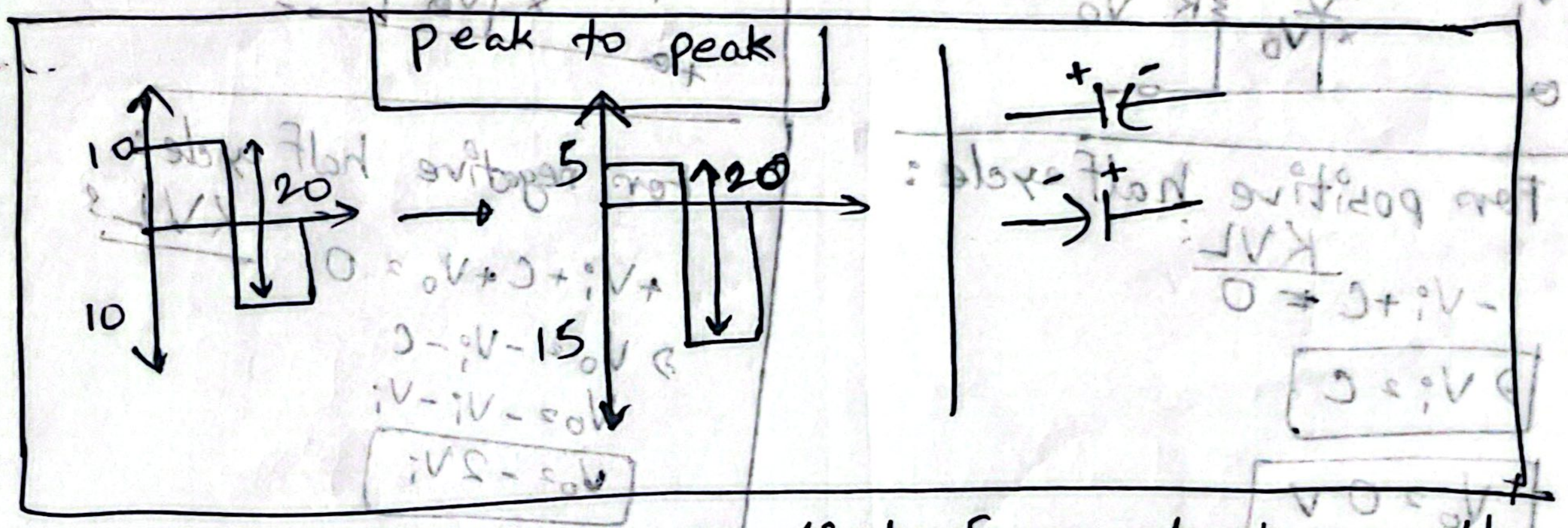
Pg-88

Clamper

No কাটাকাটি, only shifting voltage levels

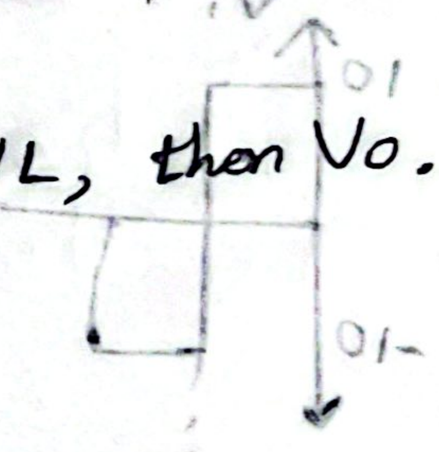


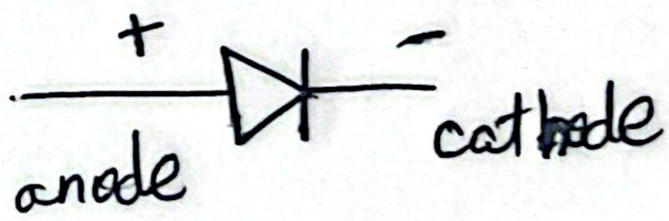
→ first cycle, make the diode forward, charge the capacitor.

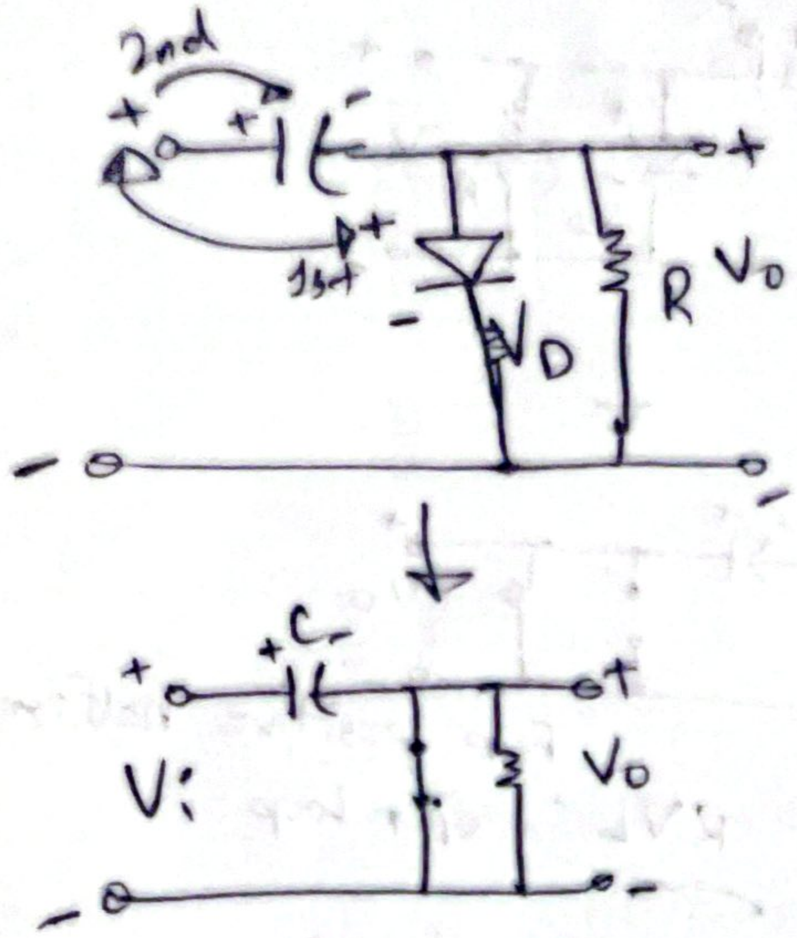
→ Use KVL in positive & negative half cycle.

→ Find Capacitor Voltage for First KVL, then V_o.

→ Then V_o for 2nd KVL.





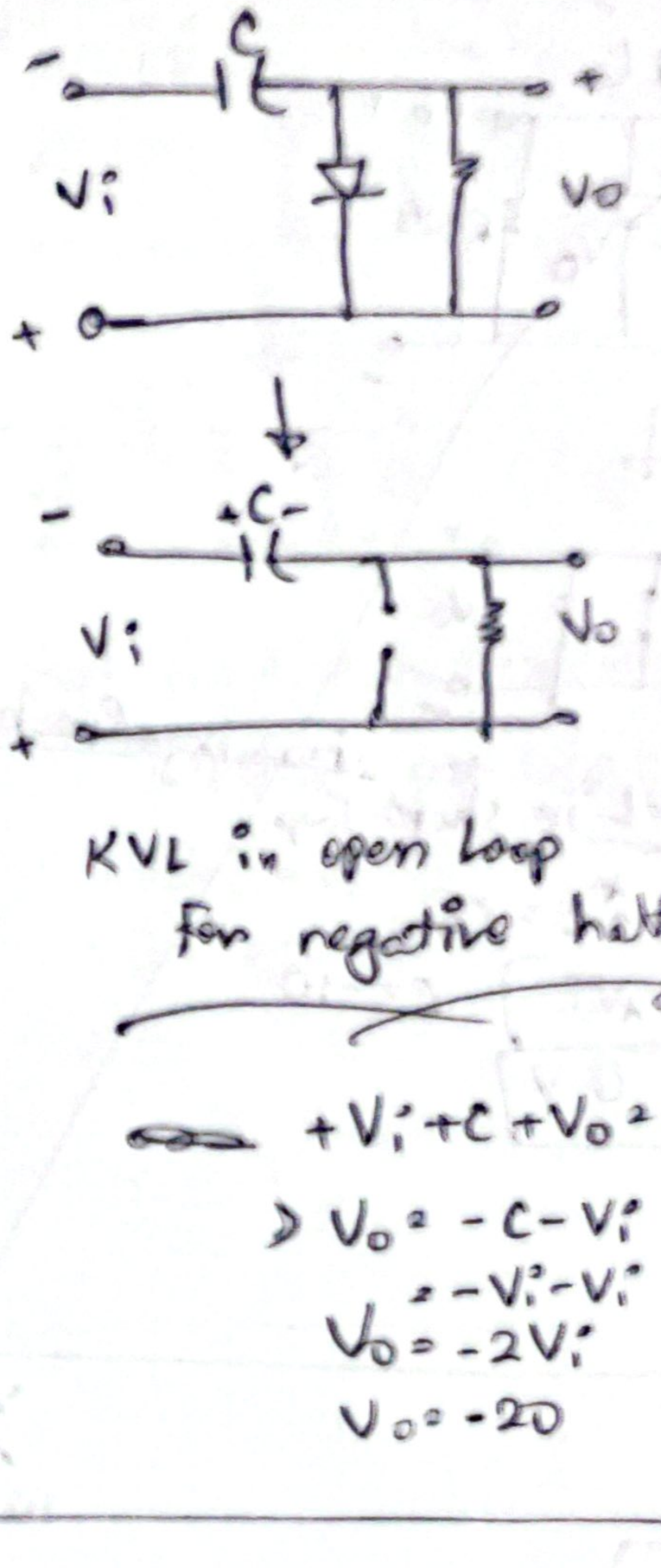


KVL in short loop
for positive half cycle

$-V_i + C = 0$

$C = V_i$ $\quad C = 10$

$V_0 = 0V$



KVL in open loop
for negative half ~~half~~ cycle

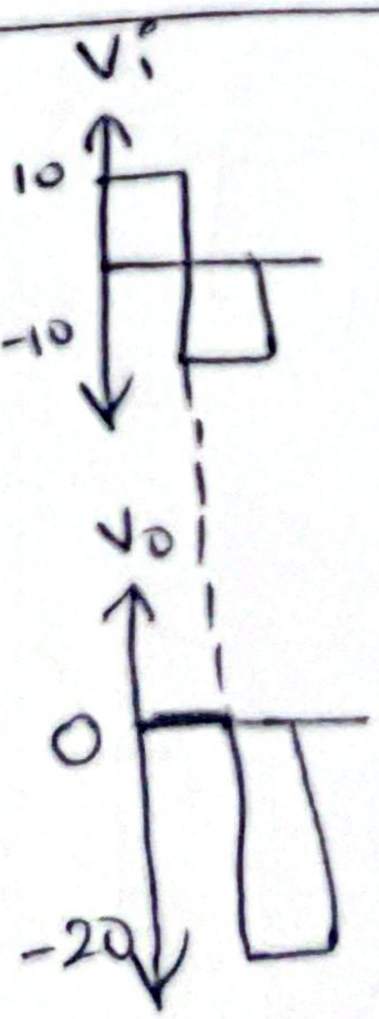
$+V_i + C + V_0 = 0$

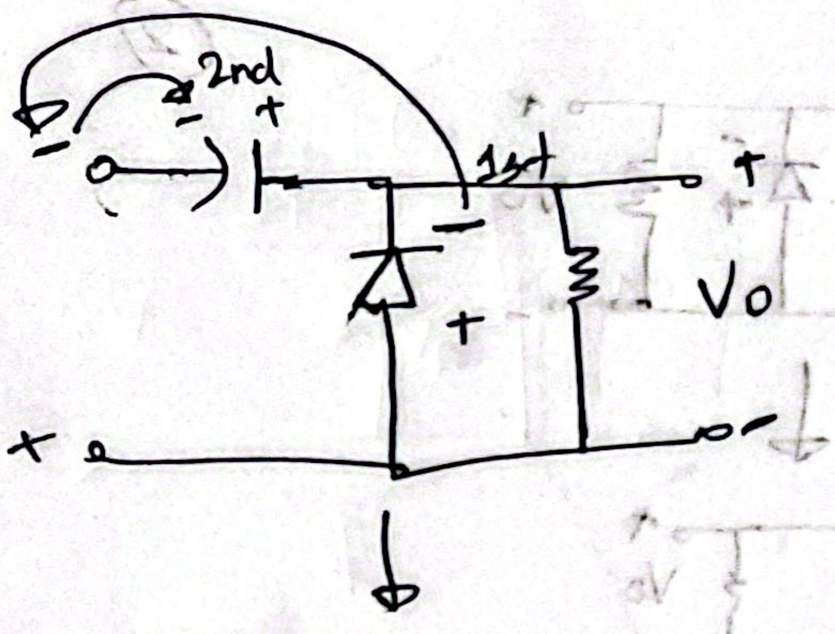
$V_0 = -C - V_i$

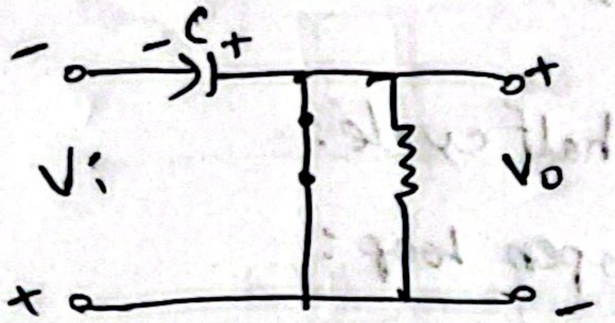
$= -V_i - V_i$

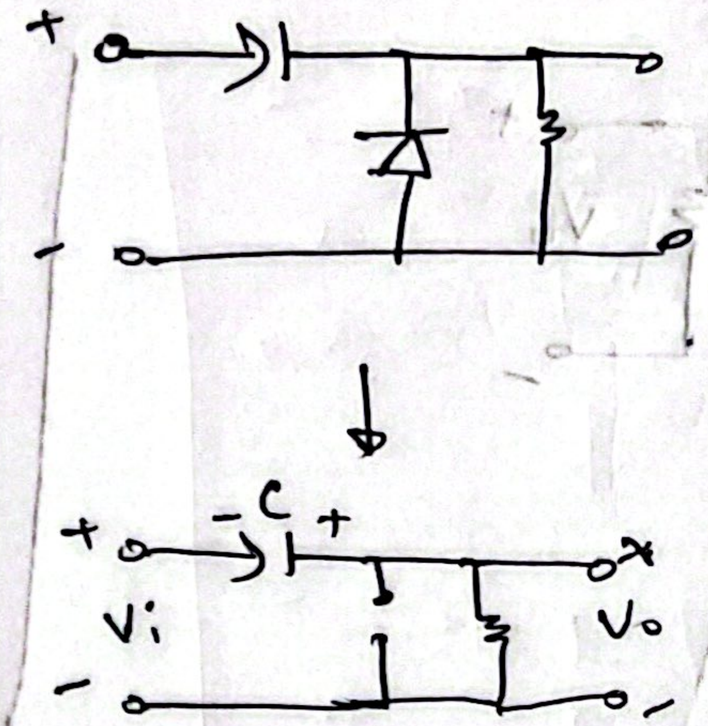
$V_0 = -2V_i$

$V_0 = -20$









KVL for short loop
in negative half cycle

$+V_i - C = 0$

$\Rightarrow C = V_i$ $\quad C = 10$

$V_o = 0V$

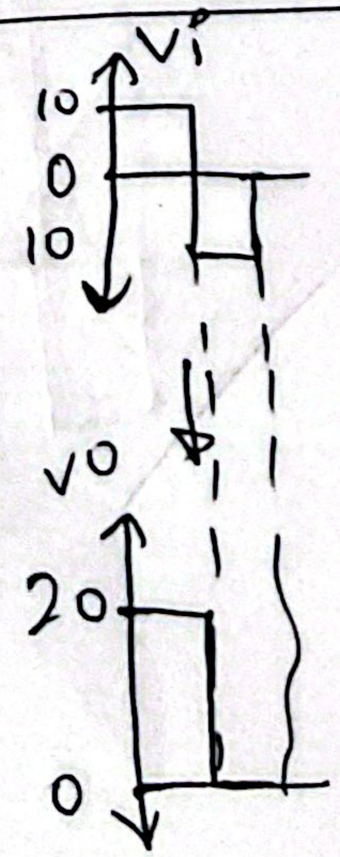
KVL for open loop
in positive half cycle

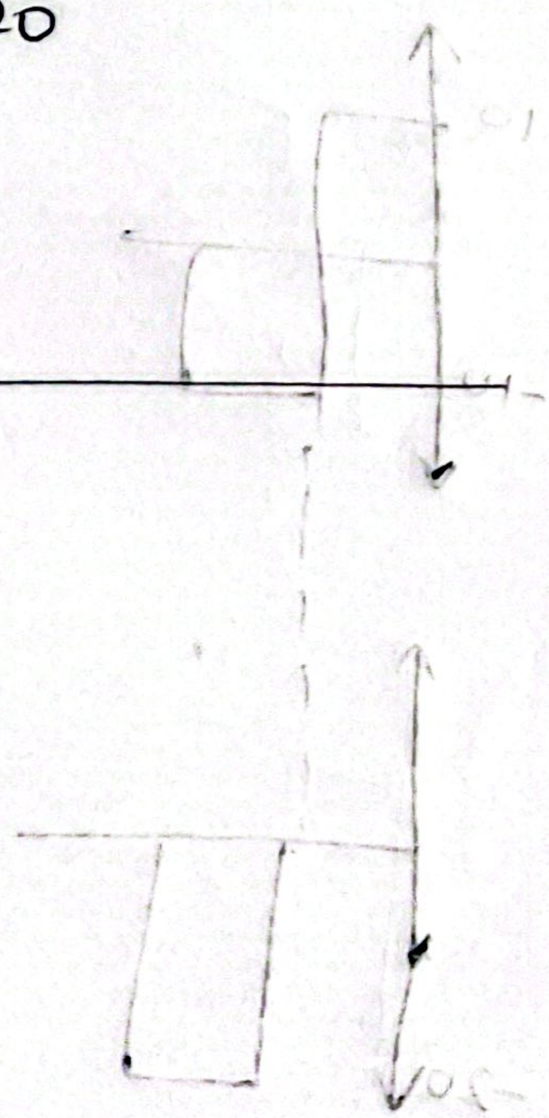
$-V_i - C + V_o = 0$

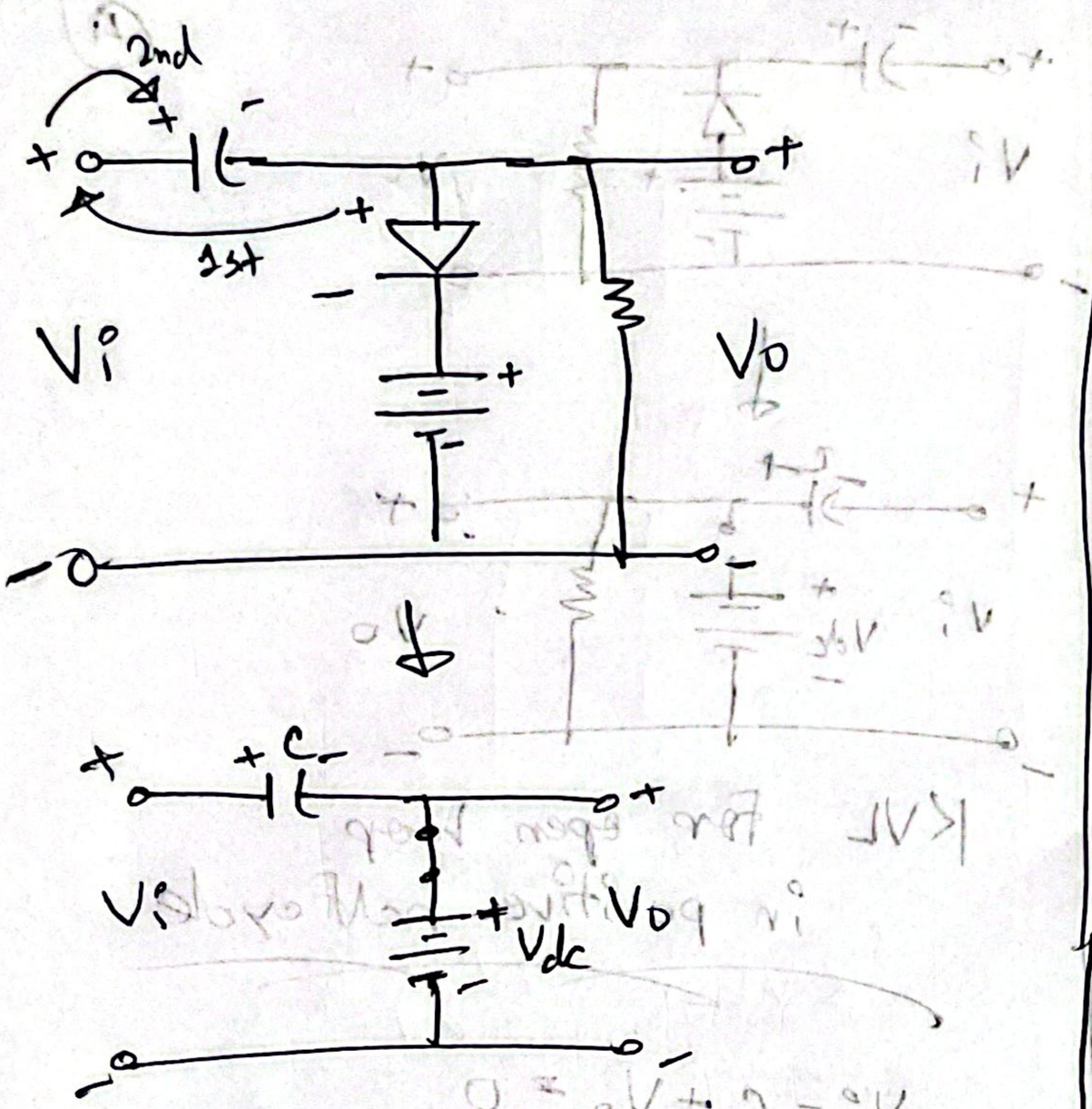
$\Rightarrow V_o = C + V_i$

$V_o = 2V_i$

$V_o = 20$







KVL for short loop in positive half cycle

$-V_i + C + V_{dc} = 0$

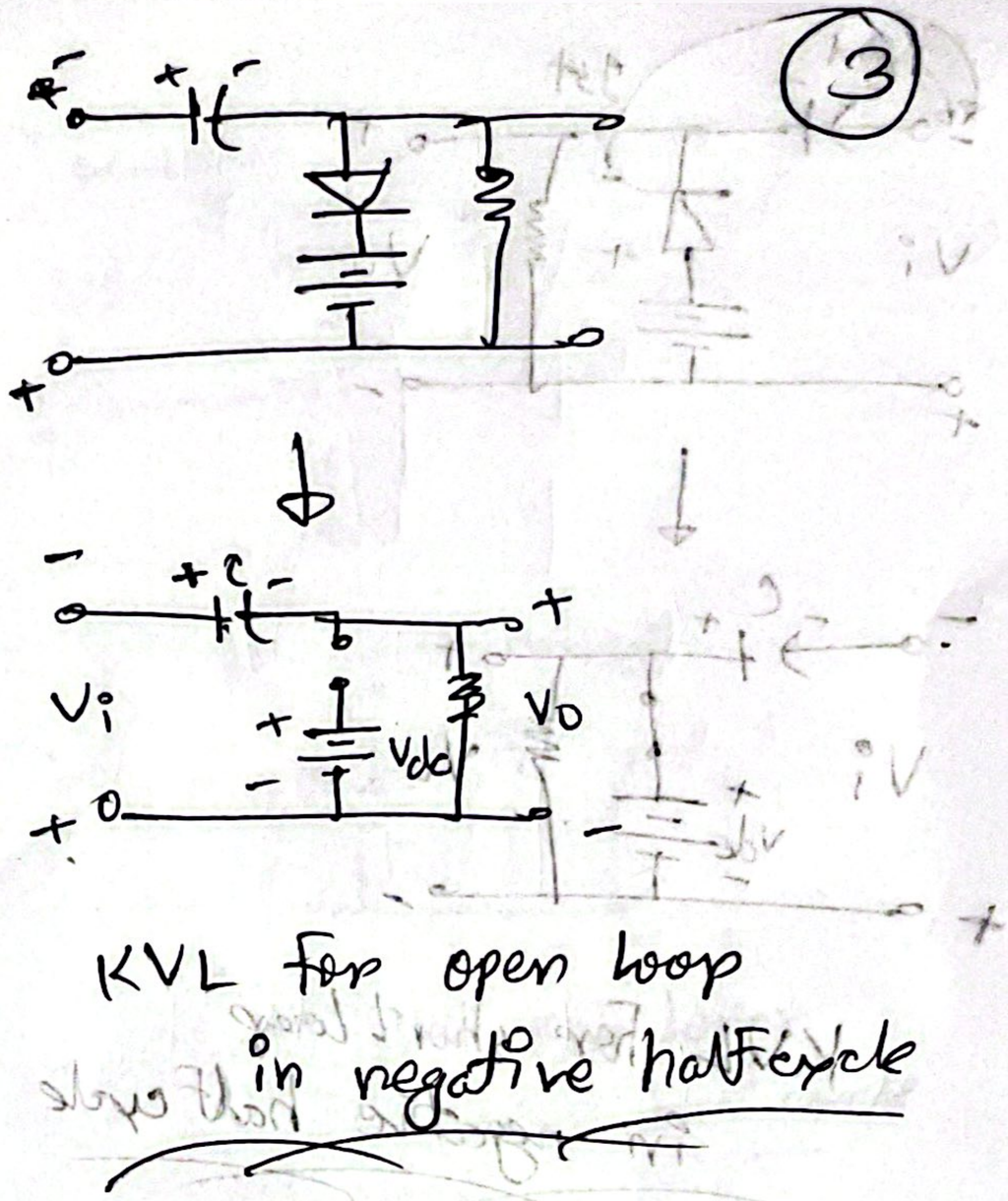
$\therefore C = V_i - V_{dc}$

$= 10 - 5$

$C = 5$

$V_o = V_{dc}$ V

$V_o = +5$ V

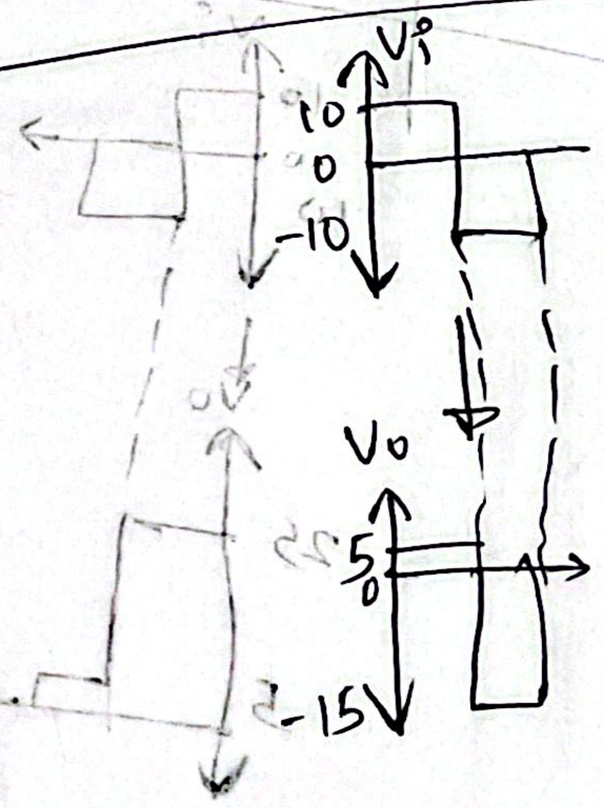


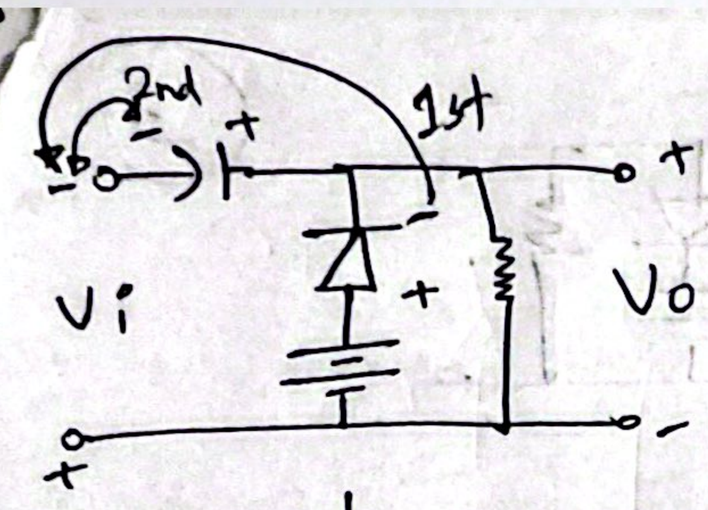
KVL for open loop in negative half cycle

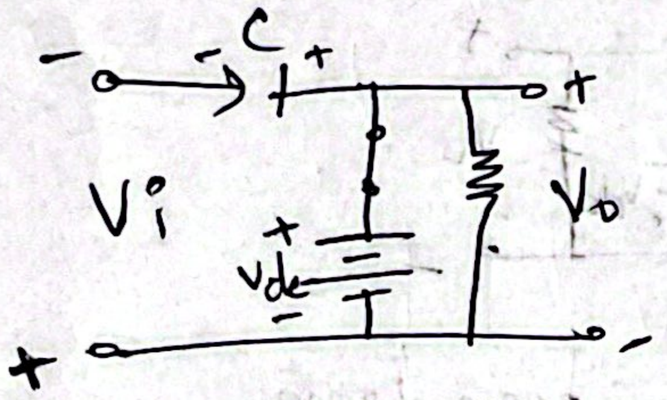
$+V_i + C + V_o = 0$

$\therefore V_o = -(V_i + C)$

$= -15$





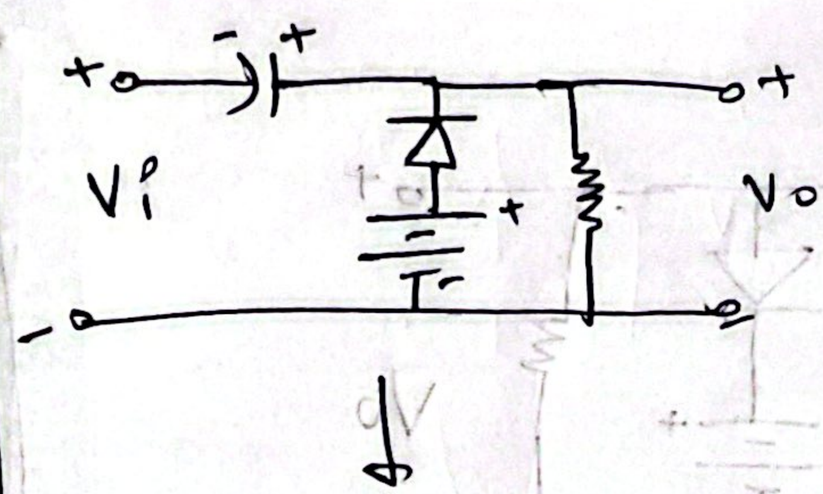


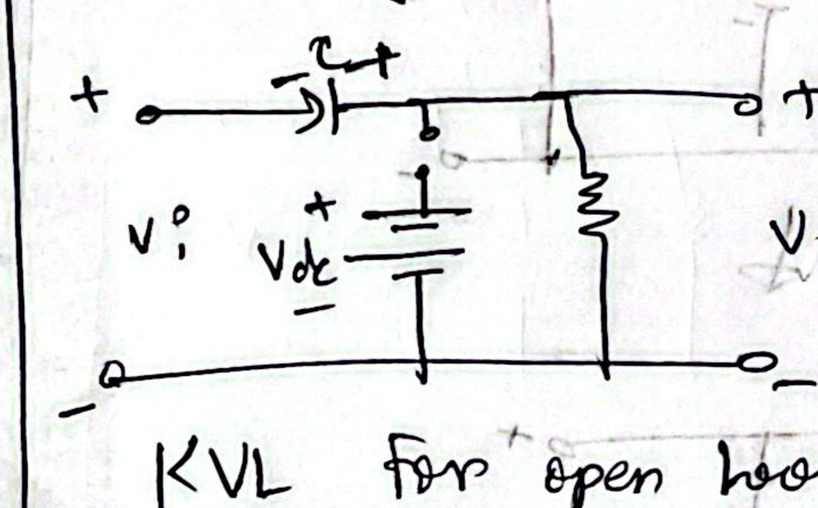
KVL for short loop
in negative halfcycle

$+V_i - C + V_{dc} = 0$

$\Rightarrow C = +(V_{dc} + V_i)$
$= +(5+10)$
$C = +15$

$V_o = +V_{dc}$
$V_o = +5V$

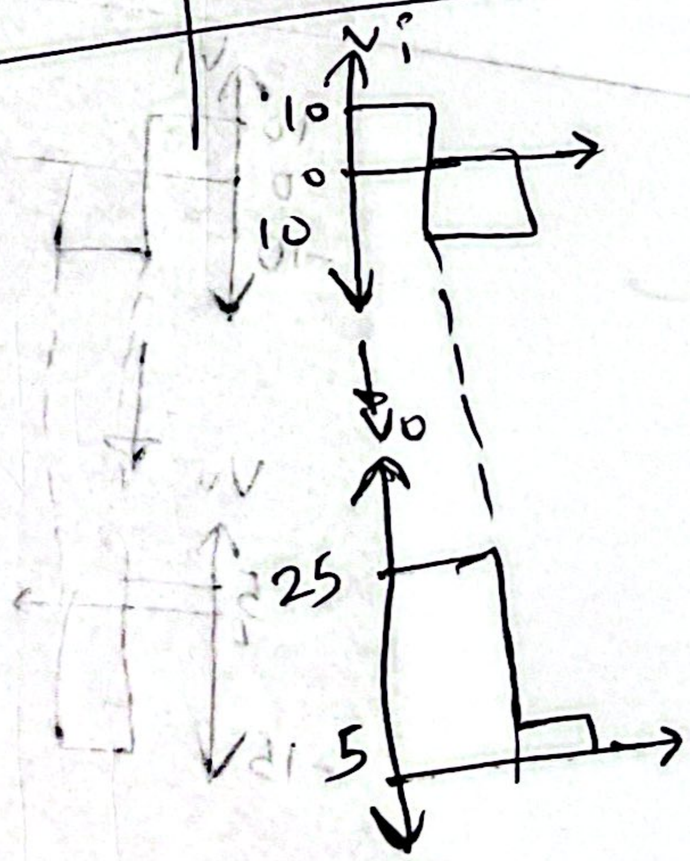


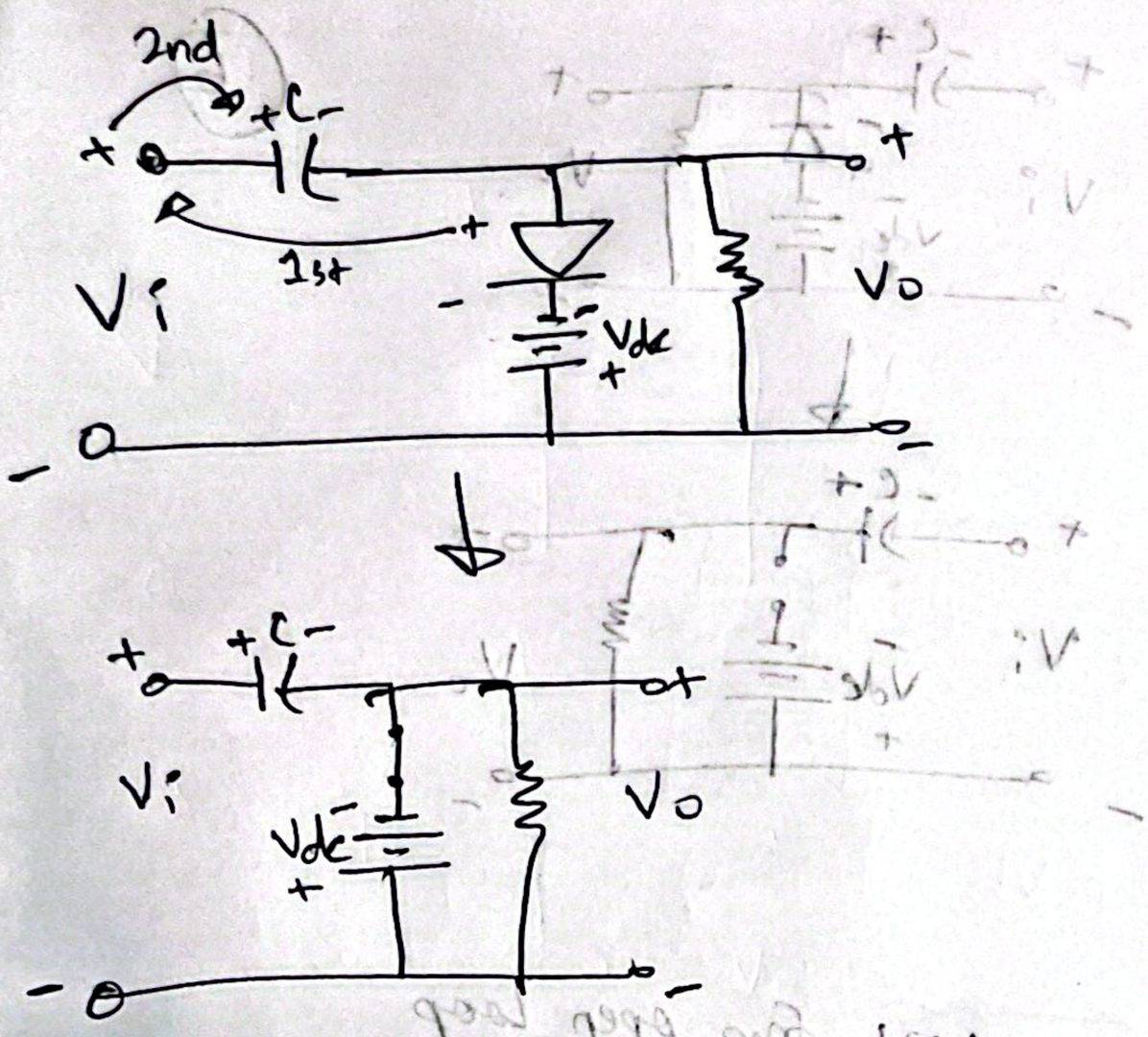


KVL for open loop
in positive halfcycle

$-V_i - C + V_o = 0$

$\Rightarrow V_o = C + V_i$
$= +15 + 10$
$V_o = +25\ V$





KVL For short loop in positive half cycle

$-V_i + C - V_{dc} = 0$

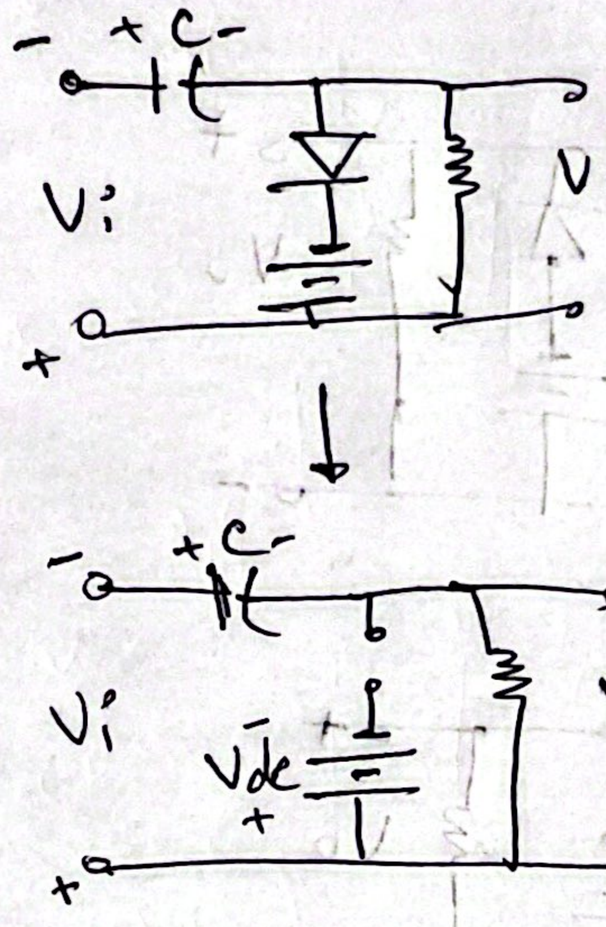
$\Rightarrow C = V_i + V_{dc}$

$= 10 + 5$

$C = 15\,V$

$V_o = V_{dc}$

$= 5V$

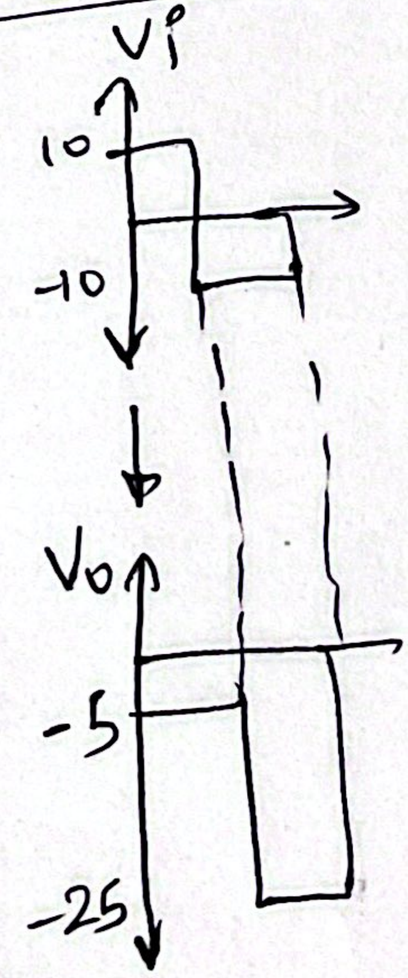


KVL For open loop in negative half cycle

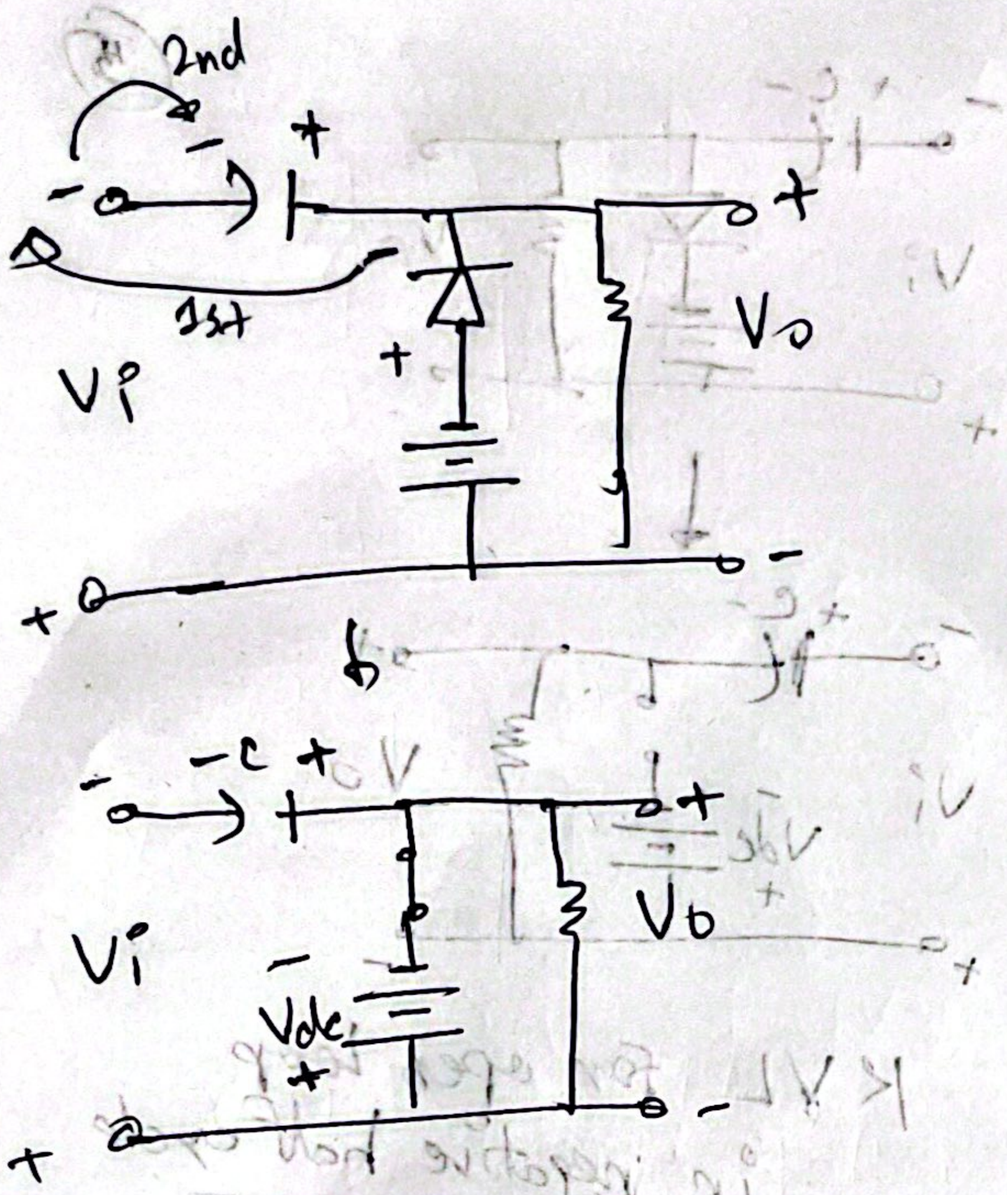
$+V_i + C + V_o = 0$

$\Rightarrow V_o = -(C + V_i)$

$= -25V$





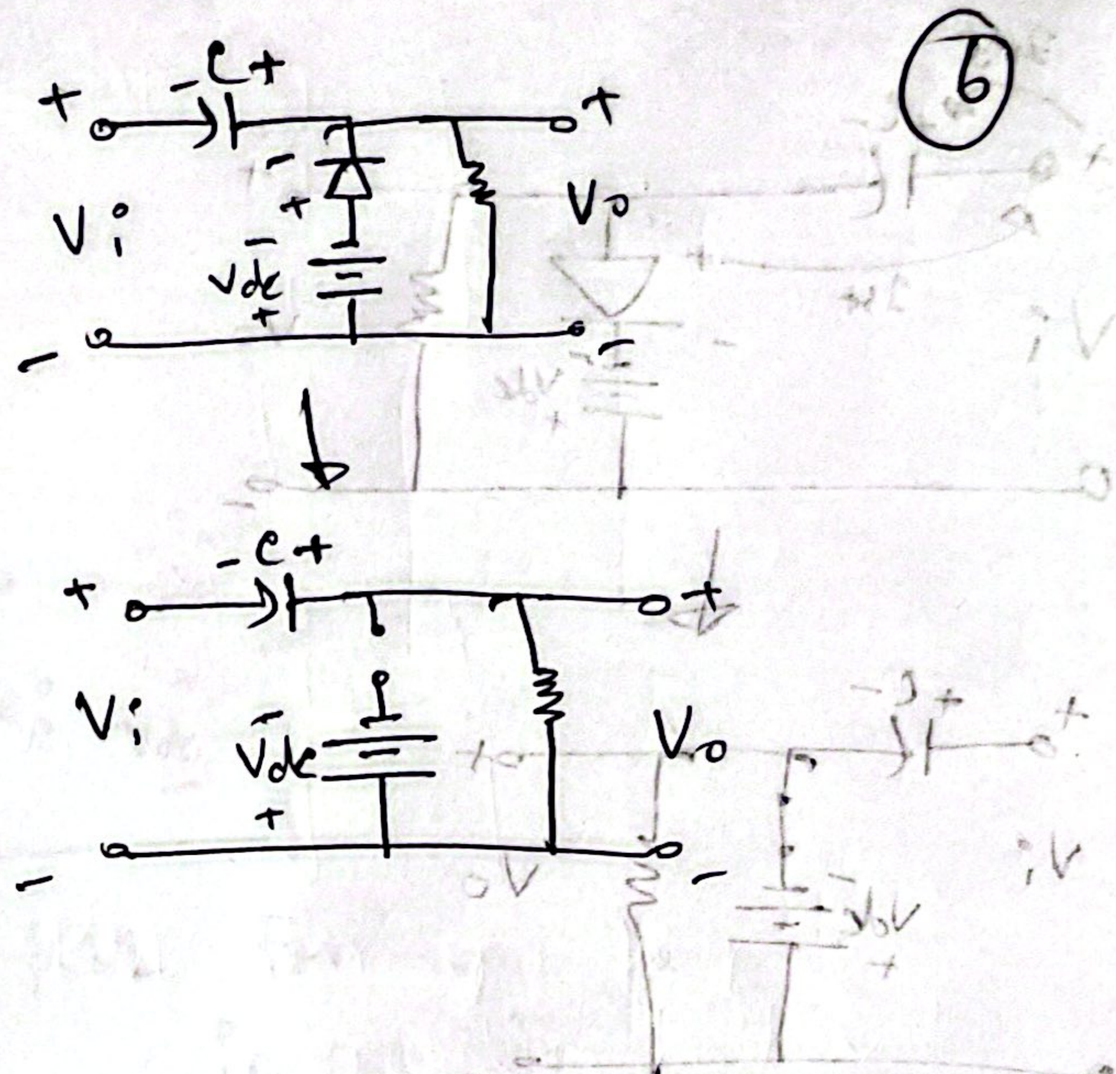


KVL for short loop in negative half cycle

$+V_i - C - V_{dc} = 0$

$\Rightarrow C = V_i - V_{dc}$
$= 10 - 5$
$= 5\ V$

$V_o = -V_{dc}$
$= -5\ V$



KVL for open loop in positive half cycle

$-V_i - C + V_o = 0$

$\Rightarrow V_o = C + V_i$
$= 15\ V$

